

Algebra 1
Unit 12
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

A study shows that temperature has a negative correlation with hot chocolate sales.

1. What does it mean that there is a negative correlation?

As the temperature increases, hot chocolate sales decrease.

2. Does this mean that falling temperatures causes more hot chocolate to be sold?

No, correlation does not mean causation. There is just a relationship between the two variables.

3. What other factors might contribute to this association?

Hot chocolate may be more readily available and sold in falling temperatures but isn't necessarily offered or sold in warmer temperatures.

4. What other information would be helpful to determine whether falling temperatures causes hot chocolate sales to rise?

Prices, specials, availability

At a theme park, there is a positive correlation between the temperature increasing and the number of lunch time restaurant reservations.

5. What does it mean that there is a positive correlation?

As the temperature increases, the number of lunch reservations increase.

6. Does this mean that increased temperature leads to more lunch reservations being made?

No, just correlated but not necessarily the cause of increased reservations.

7. What other factors might contribute to this association?

Other weather factors, park attendance

At a water park, as rain increases, attendance decreases.

8. What kind of correlation does this association represent?

Negative

9. Is there evidence of a causal relationship?

No

10. What other information would be helpful to evaluate the claim of a causal relationship?

It would need to be a controlled study or experiment.